CS541 Challenge Report

Machine learning image classification challenge for CS541 (Applied Machine Learning), comparing a custom CNN, a fine-tuned pretrained CNN, and a fine-tuned Vision Transformer for robust object recognition.

Part 1: Custom CNN from Scratch

Architecture

A 5-block CNN called `DeepCNN`. The architecture entail:

- **Stem**: 3-64 channels, 3×3 conv + BatchNorm + ReLU
- **Stage 1**: 2 residual blocks (64-64, stride=1)
- **Stage 2**: 2 residual blocks (64-128, stride=2)
- **Stage 3**: 2 residual blocks (128-256, stride=2)
- **Stage 4**: 2 residual blocks (256-512, stride=2)
- **Head**: Global Average Pooling -> Dropout(0.4) -> Linear(512, 100)

Each ResidualBlock contains two conv layers with BatchNorm and ReLU, plus a skip connection.

Hyperparameters

Parameter	Value
Optimizer	SGD + Nesterov momentum
Learning rate	0.1
Momentum	0.9
Weight decay	5e-4
Epochs	50

Parameter	Value
Batch size	128
LR schedule	Cosine Annealing
Loss	Cross Entropy Loss (label_smoothing=0.1)

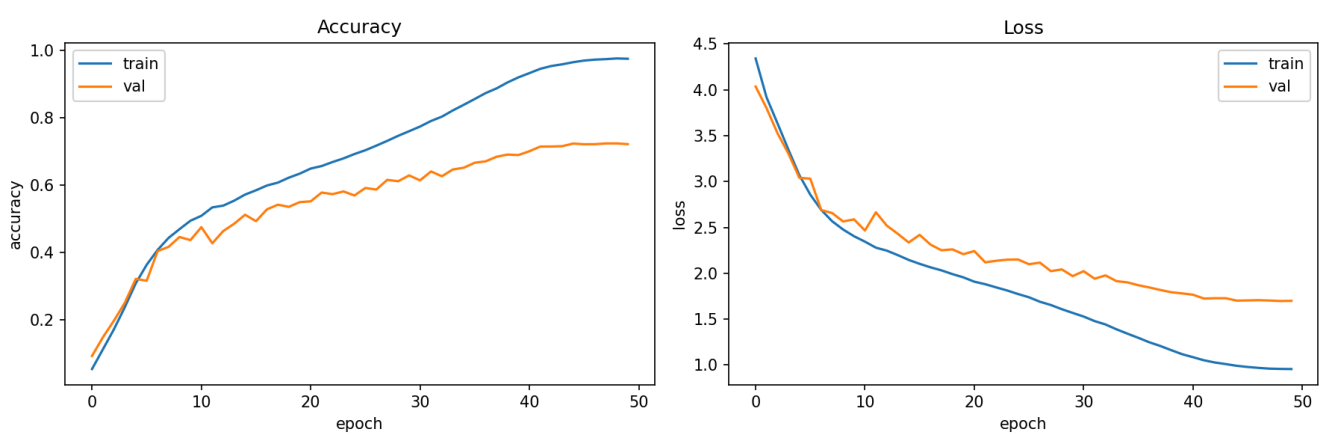
Regularization

- **Label smoothing (0.1)**: prevents overconfident predictions
- **RandomCrop (32, padding=4)**
- **RandomHorizontalFlip**: left-right flipping with 50% probability
- **ColorJitter**
- **RandomErasing (p=0.25)**: randomly masks out patches

Results

- Best validation accuracy: **72.15%**
- Clean test accuracy: **73.63%**

Loss and Accuracy plots



What Worked

SGD with Nesterov momentum and CosineAnnealingLR converged reliably. The residual connections were also critical so without them, training a would've brought about vanishing gradients. Global Average Pooling significantly reduced parameters compared to flattening.

What Didn't Work

The model showed a growing gap between train and val accuracy after epoch 30, indicating some overfitting even with regularization. Increasing dropout or adding more augmentation maybe could help.

What I Learned

That building from scratch has design tradeoffs and paying attention to every architectural choice directly as it affects both accuracy and training stability.

Part 2: Fine-Tuning a Pretrained CNN (ResNet-50)

Architecture

ResNet-50 pretrained on ImageNet with the final FC layer replaced by a 2-layer MLP head: `Linear(2048, 512) -> ReLU -> Dropout(0.3) -> Linear(512, 100)`

Input images were resized to 64×64 as a middle ground between CIFAR's 32×32 and ImageNet's 224×224.

Hyperparameters

Parameter	Phase 1	Phase 2
Optimizer	Adam	AdamW
Learning rate	1e-3	1e-4
Epochs	5	15
Backbone	Frozen	Unfrozen
Weight decay	1e-4	1e-4
Loss	CrossEntropyLoss (label_smoothing=0.1)	same

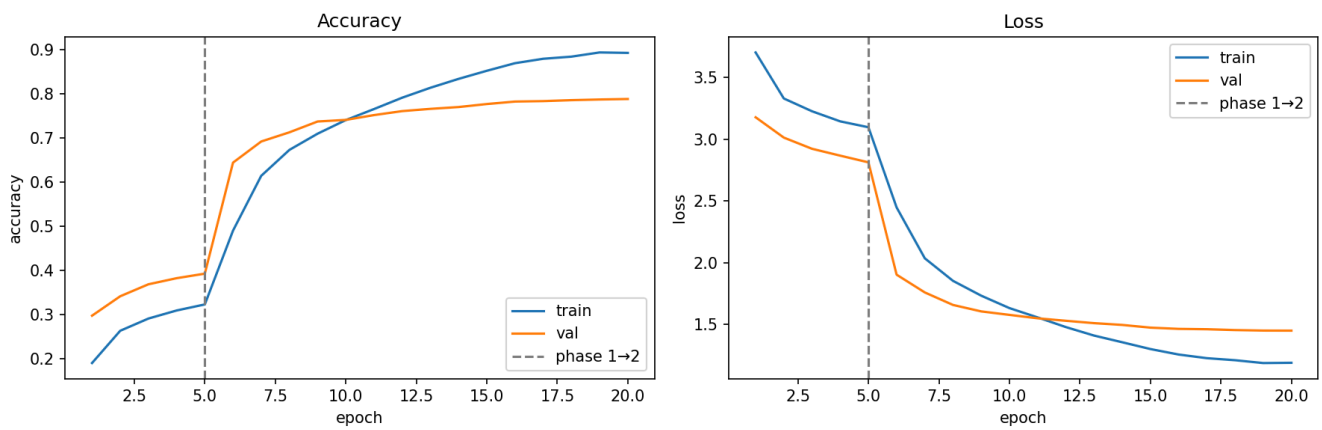
Regularization

- Label smoothing (0.1)
- RandomCrop, RandomHorizontalFlip, ColorJitter
- RandomErasing (p=0.25)

Results

- Best validation accuracy: **78.19%**

Loss and Accuracy Plots



What Worked

The two-phase strategy was very effective. Freezing the backbone in phase 1 let the new head stabilize before the pretrained weights were touched, which prevented early instability. The jump from phase 1 to phase 2 val accuracy was significant (~40% -> 78%).

What Didn't Work

ResNet-50 at 64×64 input still underperforms ViT at 224×224. Upscaling could improve results but was too slow.

What I Learned

Fine-tuning is really more efficient than training from scratch. ResNet-50 hit 78% val accuracy in 20 total epochs compared to 72% for the custom CNN in 50 epochs.

Part 3: Fine-Tuning a Pretrained Transformer (ViT-B/16)

Architecture

ViT-B/16 pretrained on ImageNet. The model divides each 224×224 image into 196 patches of 16×16 pixels, processes them as a sequence using multi-head self attention, and produces a classification token. The original head (1000 ImageNet classes) was replaced with: `Linear(768, 512) -> Relu -> Dropout(0.3) -> Linear(512, 100)`

Hyperparameters

Parameter	Phase 1	Phase 2
Optimizer	Adam	AdamW
Learning rate	1e-3	1e-4
Epochs	5	20
Backbone	Frozen	Unfrozen
Weight decay	1e-4	1e-4
Loss	CrossEntropyLoss (label_smoothing=0.1)	same

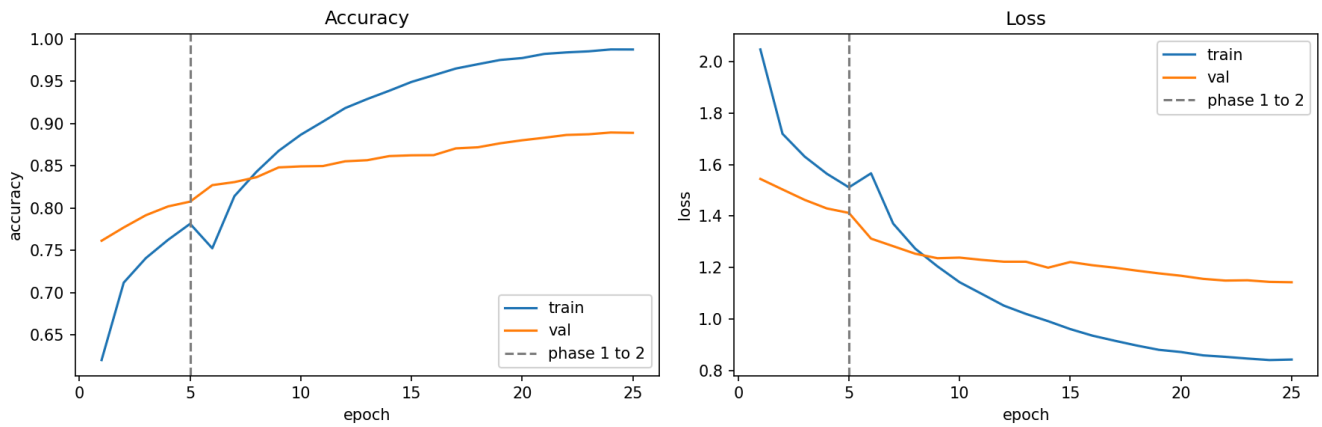
Regularization and Augmentation

- **RandomResizedCrop(224, scale=(0.7, 1.0))**
- **RandAugment(num_ops=2, magnitude=9)**: applies 2 random strong augmentations per image
- **RandomErasing (p=0.4)**: patch masking for robustness
- **RandomHorizontalFlip**
- **Label smoothing (0.1)**

Results

- Best validation accuracy: **88.86%**

Loss and Accuracy Plots



Top 3 Worst-Performing Classes

Class	Accuracy
boy	61.22%
maple_tree	68.75%
woman	69.57%

The confusion between boy/woman and other human categories (man, girl) suggests the model struggles with fine-grained intra-category distinctions when visual features overlap. maple_tree is likely confused with other tree classes as well.

Top 3 Highest-Confidence Wrong Predictions

Rank	True Label	Predicted	Confidence
1	tulip	rose	97.02%
2	possum	bear	96.46%
3	squirrel	bear	96.37%

These errors show semantically adjacent classes like flowers that share similar petal shapes, and small furry animals with similar textures.

What Worked

ViT significantly performed better thn both the custom CNN and ResNet models. The global self-attention mechanism allows ViT to capture long-range spatial relationships that CNNs miss with local convolutions. RandAugment meaningfully improved OOD robustness compared to TrivialAugmentWide.

What Didn't Work

Despite strong val accuracy (88.86%), the OOD Kaggle score was 75.7%. The distortion types in the test set show severe corruptions that the model wasn't exposed to during training.

What I Learned

ViT is the strongest model for this task, but the gap between clean val accuracy and OOD accuracy shows that augmentation alone may not be sufficient to bridge the distribution shift. Test time augmentation could be needed to close that gap more.

AI Disclosure

How I used AI:

I used Claude Chat and ChatGPT for guidance on code structure, debugging assistance, reinforcing understanding for concepts, and improving my training and evaluation. AI was also used to proofread this report.

Written by me:

- Hyperparameter decisions after reviewing training results
- Integration of all three parts (CNN, ResNet, ViT) into a single cohesive notebook.
- Adjustments to the augmentation pipeline

- Implementation of evaluation for worst-performing classes and highest-confidence errors
- Further analysis of model performance and explanations, as well as visualizations

AI assistance:

- Initial structure for make_resnet_loaders, make_vit_loaders and build_vit model
- The ResizeWrapper class after debugging for the ViT inference
- Debugging for DataLoader issues, tensor shape mismatches, and training errors
- Guidance on augmentation strategies to improve performance, early stopping, and model evaluation techniques